Consequences of Foreign Direct Investment on Inflation

Sanjeev Nhemhafuki* Subin K.C.* Resham Thapa-Parajuli *

Central Department of Economics
Tribhuvan University

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Motivation

Image 1: Inflation

Motivation

- Touching the lives of ordinary citizens, posing a grave threat to any political system. [7]

Image 2: Adapted from Onlinekhabar.com

Motivation

- A hot topic for debate: individuals; businesses; governments; and scholars.

Image 3: People speaking about inflation

Motivation

- Inflationary pressures; experienced all across the globe.

Image 4: Breaking headlines of inflation

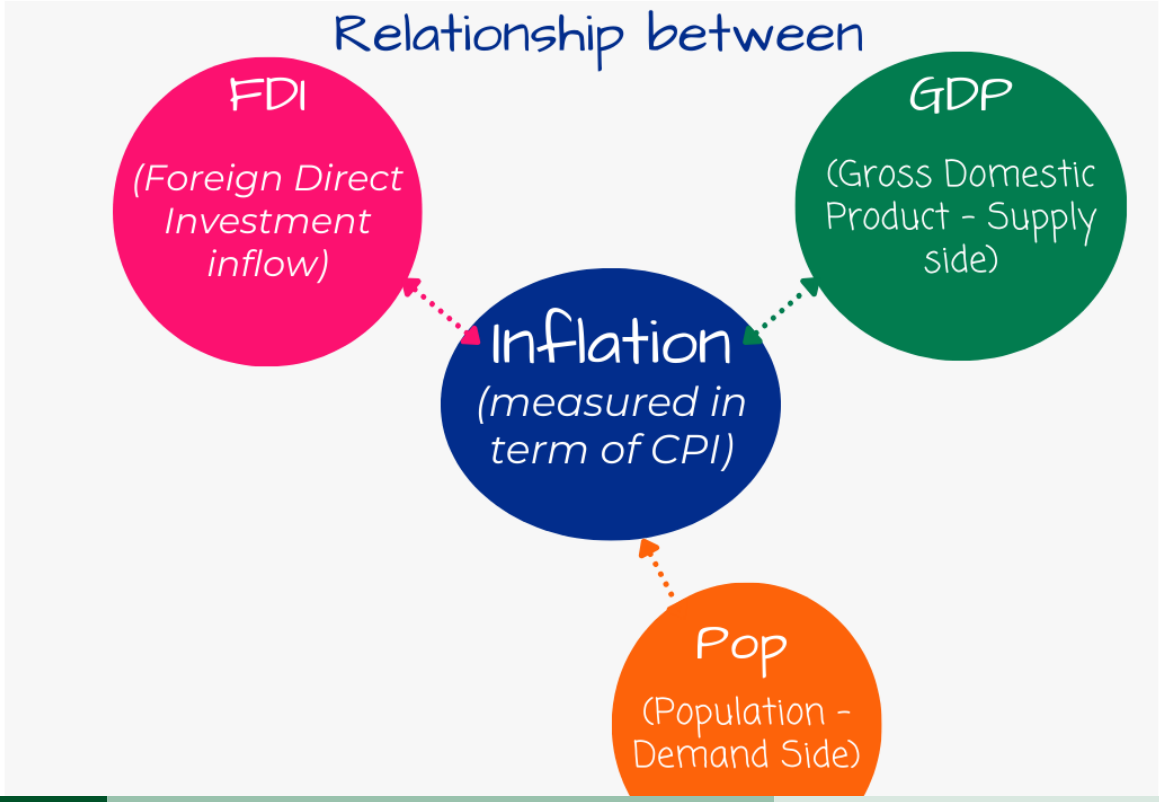
Motivation

- US, in the wake of the COVID-19 pandemic, inflation reached 8.5%, its highest rate since 1982. [8]
- The UK experienced an Annual of 8.8% CPIH highest since 1991. [13]
- Germany also has hit the record of highest inflation of 10.4% which is the highest since 1974. [14]
- All of the these countries have experienced stable price levels since the decades of 1970s.
- 1970: a decade of inflation and more precisely “stagflation”. [4]

Research Question

RQ:

- What effect does the influx of FDI have on Inflation?



Summary of the review of Literature I

- Positive Effects of FDI on Economic Growth: Sengupta and Puri (2018) [12]; Verma (2020) [17]; Asghar, N., et al. (2011) [2]; Agrawal (2015) [1]; Zekarias (2016) [18]
- Short-term vs. Long-term Relationships: Coban and Yussif (2019) [5]
- Variability in FDI-Economic Growth Links: Kaur (2013) [9] ; Basnet and Pradhan (2014) [3]
- Efficiency and Regulation: Papaioannou & Dimelis (2019) [10]

Summary of the review of Literature II

- Technology Transfer and Spillover Effects: Sur & Nandy (2018) [15]
- Inflation's Impact on FDI: Sayek (2009) [11]; Sayek (1999) [11]; Valli & Masih (2014) [16]; Coban & Yussif (2019) [5]
- Role of FDI in Decreasing Inflation: Hossain (2018) [6]

Research Gap

RG:

- Limited studies address FDI's effect on inflation.
- Existing research lacks diversity in country characteristics and observations.
- Fill gap: extensive longitudinal/cross-country time-series data for comprehensive analysis.

Data I

Sources of Data

- World Bank Data

Structure of data

- Longitudinal/Cross-country time-series

Period under Study

- 1970 to 2021

Data II

Countries of study: 88 Countries

Algeria	Burundi	Denmark	Guatemala	Jamaica	Morocco	Peru	Sweden
Australia	Cambodia	Dominican Republic	Guinea-Bissau	Japan	Nepal	Portugal	Thailand
Austria	Cameroon	Ecuador	Honduras	Kenya	Netherlands	Rwanda	Togo
Bangladesh	Canada	Egypt, Arab Rep.	Hong Kong SAR, China	Korea, Rep.	New Zealand	Saudi Arabia	Tonga
Barbados	Chad	El Salvador	Iceland	Lesotho	Nicaragua	Senegal	Tunisia
Belgium	Chile	Finland	India	Madagascar	Niger	Sierra Leone	Turkiye
Benin	Colombia	France	Indonesia	Malaysia	Nigeria	Singapore	Uganda
Bolivia	Congo, Rep.	Gambia, The	Iran, Islamic Rep.	Mali	Norway	South Africa	United Kingdom
Botswana	Costa Rica	Germany	Iraq	Malta	Oman	Spain	United States
Brazil	Cote d'Ivoire	Ghana	Ireland	Mauritania	Pakistan	Sri Lanka	Uruguay
Brunei Darussalam	Cyprus	Greece	Italy	Mexico	Paraguay	Sudan	Zimbabwe

Data III

Descriptive Statistics

Variable	Mean	Std. dev.	Min.	Max.	Observations
Inf overall	1.701	1.213	-4.791	9.372	N=3627
between		0.592	0.801	2.865	n=52
within		1.066	-3.894	8.893	T-bar=69.75
FDI overall	19.929	2.93	6.908	27.322	N=3627
between		1.51	17.251	21.802	n=52
within		2.529	7.219	26.163	T-bar=69.75
GDP overall	24.376	2.278	17.346	30.78	N=3627
between		0.809	22.592	25.658	n=52
within		2.14	18.337	20.046	T-bar=69.75
Pop overall	16.382	1.64	11.471	21.065	N=3627
between		0.208	16.017	16.908	n=52
within		1.628	11.185	20.892	T-bar=69.75

Data IV

Pairwise Correlation Matrix

	GDP	FDI	Pop	Inf
GDP	1.000			
FDI	0.830***	1.000		
	(0.000)			
Pop	0.698***	0.439***	1.000	
	(0.000)	(0.000)		
Inf	-0.246***	-0.345***	0.063**	1.000
	(0.000)	(0.000)	(0.001)	

Methods I

Model Specification

The specified model is as follows:

Inf = f (FDI, GDP, Popn) ..... (1)

Where,
Inf = Inflation (measured in terms of CPI)
FDI = Foreign Direct Investment
GDP = Gross Domestic Product
Popn = Population

The model is presented as follows:

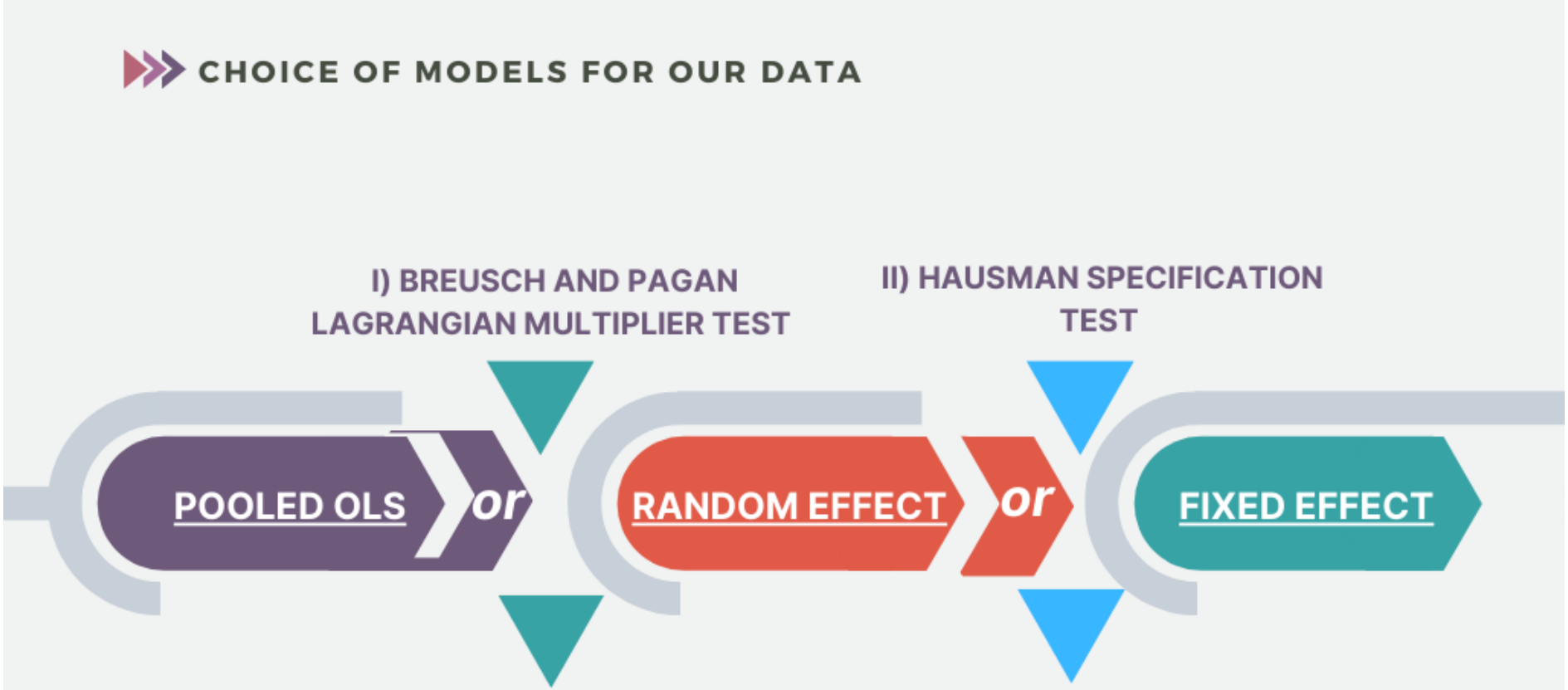
$$INF_{(i,t)} = \alpha_0 + \beta_1 FDI_{i,t} + \beta_2 GDP_{i,t} + \beta_3 Popn_{i,t} + \mu_{i,t}$$

Methods II

Methodology

- Regression: OLS, Random Effect, Fixed Effect
- Breusch-Pagan Lagrange multiplier (LM) test (Breusch and Pagan 1980): Random effects or OLS Model.
- Hausman Specification Test (Hausman 1978): Fixed effect model or Random effect model.

Methods III


CHOICE OF MODELS FOR OUR DATA
I) BREUSCH AND PAGAN LAGRANGIAN MULTIPLIER TEST
II) HAUSMAN SPECIFICATION TEST
POOLED OLS
or
RANDOM EFFECT
or
FIXED EFFECT
• Panel data not poolable.
• chibar^2(01) = 5739.32
• Prob > chibar2 = 0.0000
• Fixed>Random
• chi^2 = 121.781
• P-value = 0.0000

Results

Regression Results

Dependent Variable: Inf (Inflation: measured in terms of CPI)

Models & VARIABLES	Pooled OLS Model		Fixed Effect Model		Random Effect Model	
	(1)	(2)	(3)	(4)	(5)	(6)
FDI	-0.143***	-0.115***	-0.084***	-0.016	-0.093***	-0.041***
	(0.007)	(0.012)	(0.007)	(0.012)	(0.007)	(0.012)
GDP		-0.150***		-0.228***		-0.209***
		(0.019)		(0.018)		(0.018)
Popn		0.283***		0.307***		0.302***
		(0.017)		(0.015)		(0.015)
Constant	4.544***	3.027***	3.382***	2.549***	3.579***	2.689***
	(0.130)	(0.211)	(0.139)	(0.197)	(0.144)	(0.202)
Observations	3,627	3,627	3,627	3,627	3,627	3,627
R-squared	0.119	0.19	0.119	0.172	0.119	0.181
Number of year	52	52	52	52	52	52

Note: The statistical test values are in parentheses. **, *** indicate significance at 5%, 1% levels respectively.

Discussion

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- FDI & Inflation negative relationship in Columns (1), (3) and (5) for all 3 models.
- Columns (2), (4), and (6), regression result after introducing the control variables (i.e. GDP and Population).
- Negative relationship with the outcome variable, even after controlling for control variables.
- Outcome variable loses its significance after the control variables were considered in the model (4).
- Control variables in (2), (4) and (6) have a high level of significance with the Inflation.

Conclusions

Conclusions

- Negative association to FDI to the General Price Level.
- Increase in the FDI inflow and GDP (Supply), the price level decreases.
- Increase in the Population (Demand), price level increases.
- Findings correspond to the basic microeconomic foundations of the economic theories which establish the relationship between inflation and FDI.
- Coordinated Fiscal and Monetary Policies - Encourage diversified FDI inflow.
- FDI policies - adequate assessment of countries' needs and priorities, FDI and its consequences on inflation, other macroeconomic variables, and the environment.

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THANK YOU